

Network Science

Network Models

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Chapter 4: Network Models

- Random networks
- Small-world networks
- The configuration model
- Preferential attachment
- Other preferential models

Features of real networks (1/4): *Small-world* property

Most real-world networks have **short paths**

Network	Nodes (N)	Links (L)	Average path length ($\langle \ell \rangle$)	Clustering coefficient (C)
Facebook Northwestern Univ.	10,567	488,337	2.7	0.24
IMDB movies and stars	563,443	921,160	12.1	0
IMDB co-stars	252,999	1,015,187	6.8	0.67
Twitter US politics	18,470	48,365	5.6	0.03
Enron Email	87,273	321,918	3.6	0.12
Wikipedia math	15,220	194,103	3.9	0.31
Internet routers	190,914	607,610	7.0	0.16
US air transportation	546	2,781	3.2	0.49
World air transportation	3,179	18,617	4.0	0.49
Yeast protein interactions	1,870	2,277	6.8	0.07
C. elegans brain	297	2,345	4.0	0.29
Everglades ecological food web	69	916	2.2	0.55

Note

Here, *small-world* denotes the presence of short paths. In this lecture, we will formally introduce the notion of *small-world* networks, which is not purely about having short paths.

Features of real networks (2/4): high *clustering coefficient*

Many networks have high *clustering coefficients*.

Other networks, e.g., bipartite and tree-like networks, have low clustering coefficients.

Network	Nodes (N)	Links (L)	Average path length ($\langle \ell \rangle$)	Clustering coefficient (C)
Facebook Northwestern Univ.	10,567	488,337	2.7	0.24
IMDB movies and stars	563,443	921,160	12.1	0
IMDB co-stars	252,999	1,015,187	6.8	0.67
Twitter US politics	18,470	48,365	5.6	0.03
Enron Email	87,273	321,918	3.6	0.12
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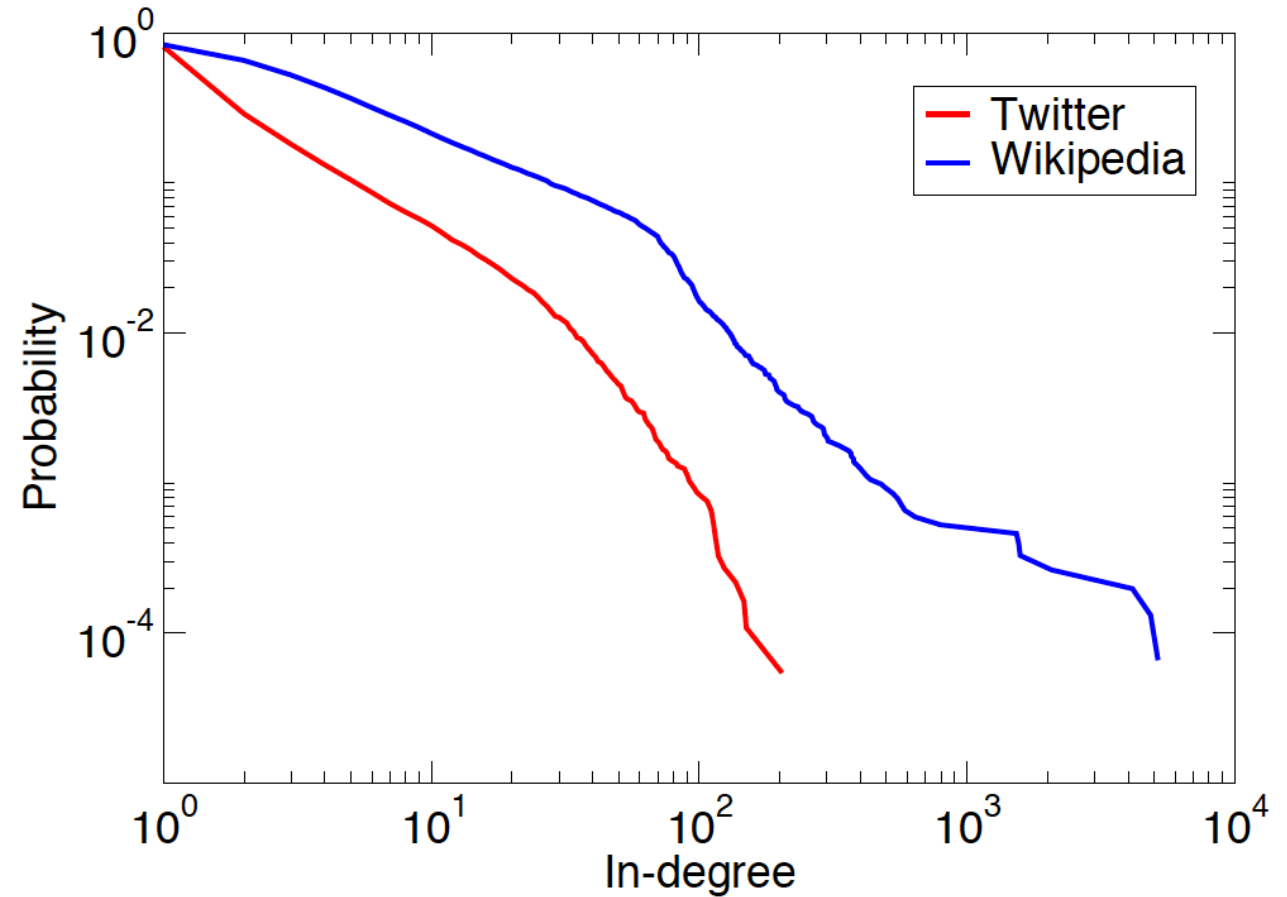
Note

For memory, the clustering coefficient of a node is the fraction of pairs of the node's neighbours that are connected to each other

Features of real networks (3/4): *heterogeneity*

Heavy-tail distributions: the variable goes from small to large values

Hubs: nodes with high degree



Features of real networks (4/4): *heterogeneity*

Heterogeneity parameter: a measure of how broad the degree distribution is:

$$\kappa = \frac{\langle k^2 \rangle}{\langle k \rangle^2}$$

Network	Nodes (N)	Links (L)	Average degree ($\langle k \rangle$)	Maximum degree (k_{max})	Heterogeneity parameter (κ)
Facebook Northwestern Univ.	10,567	488,337	92.4	2,105	1.8
IMDB movies and stars	563,443	921,160	3.3	800	5.4
IMDB co-stars	252,999	1,015,187	8.0	456	4.6
Twitter US politics	18,470	48,365	2.6	204	8.3
Enron Email	36,692	367,662	10.0	1,383	14.0
Wikipedia math	15,220	194,103	12.8	5,171	38.2
Internet routers	190,914	607,610	6.4	1,071	6.0
US air transportation	546	2,781	10.2	153	5.3
World air transportation	3,179	18,617	11.7	246	5.5
Yeast protein interactions	1,870	2,277	2.4	56	2.7
C. elegans brain	297	2,345	7.9	134	2.7
Everglades ecological food web	69	916	13.3	63	2.2

Models

Model: set of instructions to build networks

Goal: find models that generate networks with the same characteristics as real-world networks

Random networks

Simple idea: placing links at random between pairs of nodes

Algorithm (Gilbert random network model [1])

- Start with N nodes and zero links
- Go over all pairs of nodes; for each pair of nodes i and j , generate a random number r between 0 and 1
 - If $r < p \Rightarrow i$ and j get connected
 - If $r \geq p \Rightarrow i$ and j remain disconnected

Warning

NetworkX says that `gnp_random_graph` returns a random graph, also known as an Erdős-Rényi random graph [2] or a binomial graph, but this is not quite accurate. For large graphs, it is equivalent. The NetworkX function for an ER graph is `gnm_random_graph` which specifies the number of edges. Note that for maximum confusion, the function `nx.erdos_renyi_graph` is an alias for `nx.gnp_random_graph` rather than `nx.gnm_random_graph`.